DIRAC: A Forensic Framework for Quantifying Engineering and Financial Leakage in the Era of Large Language Models

Deterministic Integrity & Risk Assessment Core (DIRAC) v1.0

Rakesh Venkat

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1 Summary

While industry leaders report that 41% of new commercial code is AI-generated , there is a significant quality issues that is emerging. Recent studies also shows that 44% of AI initiatives are being rolled back due to operational costs. Total development process has actually slowed by 19% as teams struggle to integrate "messy" AI logic.

Current metrics which are used for tracking engineering productivity lacks something very critical, the **Hidden cost of AI generated code**. While companies are focused on the velocity and cost of utilizing token by buying licenses from AI vendors, there is a complete lack of visibility in **Review Tax**.

Review Tax is the massive amount of valuable and expensive senior engineering time spent fixing, auditing and refactoring AI generated code that might appear correct but are structurally fragile. Over time this leads to delays and significant waste of capital.

Recent research from UC Berkeley showed that these benchmarks can be manipulated and do not reflect the real-world performance.

Deterministic **I**ntegrity & **R**isk **A**ssessment **C**ore **(DIRAC v1.0)** is a forensic framework designed to move beyond simple dashboard and DORA metrics.

I have analysed over **2000** open source repositories with **10,005** commits and quantified the **Information Entropy** of code changes to map it directly to financial loss.

This paper shows that there is a need to **Audit Entropy** and protect engineering decay over time.

2 1. Traditional Metrics in the AI Era

For years, framework like **Deployment Frequency, Lead Time** (DORA)has been the gold standard for velocity metrics but never designed as a quality metrics. But in the era of AI, it can be potentially be misleading. Here's how

- Multiple deployments can happen in a day but if each of them contain high-entropy code that requires extensive reviews by a senior, then velocity is an *illusion*.
- DORA does not track Review Tax, it fails to show that a single AI generated commit could potentially consume significant senior team's capacity to ensure that system remains stable after merge.

- Traditional metrics cannot see the **entropy** (disorder) inside a commit. A build will only have either a *pass* or *fail* status but

3 Archival Notice

DIRAC is now archived. Active research and development have ceased, and no further updates are planned.

This paper is retained for historical and reference purposes.

For questions about the published work, you can reach me at qubitai.in@gmail.com.